The Nero wizard is the least complicated method of burning MP3s

72 to 80 minutes of music depending upon the capacity of the CD.

■ Burn it: Once the files are chosen and you are ready to burn, choose Write from the File menu. This opens a window seeking your confirmation to burn the CD. Click on the Burn button. As soon as the process begins you will notice a progress bar while each track is being burnt on the CD. Once the burning is done the red light near the progress bar turns green indicating successful completion

There are, of course, many ways of doing this—MusicMatch Jukebox is a freeware software that does the same. Another popular alternative is Easy CD Creator. The program accepts popular audio formats such as MP3 and WAV as input. Easy CD Creator supports BURN-Proof (Buffer Under RuN error Proof) technology.

Music Match JukeBox

Converting for the PC: Ripping using MusicMatch

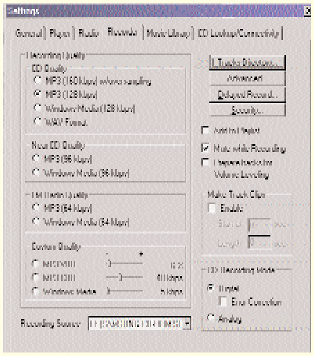
Open MusicMatch. Insert a CD into your CD-ROM. If you're not connected to the Internet, connect now. The program will automatically search the online CDDB. If it finds a match for your CD, all the artist and album information will be filled for you. If it doesn't return a match, you may need to type in the track listings.

After the CDDB returns a match or when you've completed entering artist and album names and all the track titles, hit the record button on the recorder. Depending on the speed of your PC, you should have the CD completely ripped in less than 10 minutes.

The bitrate dilemma: Which one do you pick?

Music available on CDs and other similar media can be digitally encoded into various music formats—from WMA to Ogg Vorbis to MP3. But all of them follow a common digital coding convention called bitrate. This defines how much of detail (number of bits) is to be stored per second of a song in an encoded file. Bitrate decides the quality of the encoded file and its size too. Does it really make a drastic difference in terms of sound quality when you encode files with different bitrates? The more details (more bits per second or high bit rate) you store into a file, the better the sound quality. Storing more information per second of music builds a larger MP3 file. If you are sensitive to subtle musical details and are finicky about sound reproduction, you should opt for a higher bitrate encoding (greater than 192 Kbps). 128 Kbps is the usual encoding level music enthusiasts choose. Encoding at this rate offers a perfect balance of small file size and quality of music.

Music encoding can be done in three ways—constant bits, average bits or variable bits per second of music. In constant bitrate



MP3's can be encoded with various parameters

(CBR) the same number of bits is added to each second or frame of music. Whether it's a scream or silence, the same number of bits is added into the encoded file (which is actually a waste, since periods of silence don't require more bits to be stored). Average bitrate (ABR) lets you choose an average number of bits that are to be added to each frame. Variable bitrate (VBR) is an intelligent method introduced by Xingtech, which decides on how many bits are to be put into a frame or second depending on the complexity of the sound. Software that employ the VBR encoding scheme use lesser bits for periods of silence and more bits according to the frequency (sound) level of the music.

The next time you rip your music CD, don't forget to use VBR and encode at high bitrates. You might end up with a smaller file size than if you used CBR or ABR. The only caveat is that if you are into portable MP3 players, most do not play bitrates higher than 192 Kbps. Check your player-specifications before you encode your MP3s.

Moving Music from Audio Tapes to MP3s

If you have an aging but priceless collection of tapes lying around, take a day off to try this out.

Connect the tape player to your soundcard's line-in terminal with a 0.25-inch mini-jack and record the contents of your tape as a WAV file. Nullsoft's Diskwriter plugin, or MusicMatch Jukebox or any plain vanilla wave editor will do.

Set the software to record from the line-in channel, press Play on your tape deck, and wait for the tape to play.

After the WAV file has been written, and the conversion from analog to digital is complete, comes the process of normalisation. Normalising your file involves searching for any abnormalities in it and preventing it from playing too low or too loud.

Cut the WAV file to pieces. From this you can convert to MP3 or to any other format you want using Audiocatalyst or any other encoding software. It is possible to burn the tape onto an audio CD as well, but the quality will not be as good on a CD since the original source was a tape.

Before you create the CDs, you may want to transfer the WAV files to an editing studio such as Pro Tools, SoundForge XP Studio 5.0, or CoolEdit Pro.